

PRIMARY TREATMENT OF WATER

Objective

The water from the surface source may have characteristics which are unsuitable for human consumption. Some characteristics are:

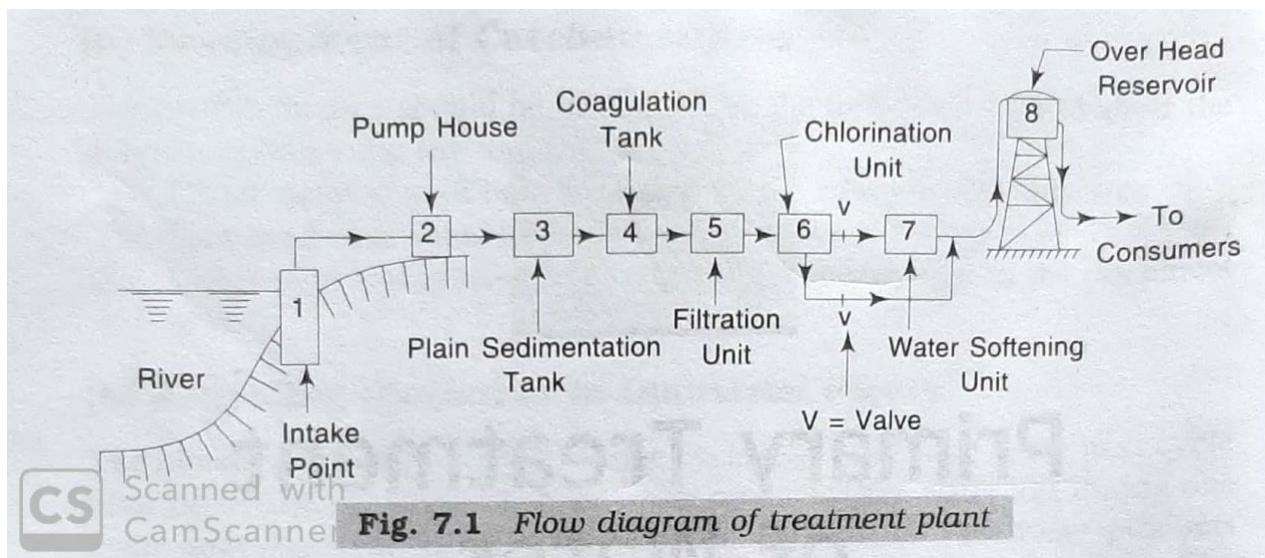
1. It may be turbid.
2. It may contain color.
3. It may contain acids, salt, and gases which have corrosive action and may impart hardness of water.
4. It may contain bacteria which may cause water borne disease.

Therefore, the object of treatment of water is to remove those impurities and make the water suitable for human consumption.

Flow diagram of a treatment unit.

Sequence of unit

1. Intake point
2. Pump house
3. Plain sedimentation tank
4. Coagulation tank
5. Filtration unit
6. Chlorination unit
7. Water softening plant
8. Overhead reservoir



Sedimentation

The particles which do not change their shape, size and weight while settling down in a fluid are known as *discrete particles*. The suspended impurities in water consist of discrete particles such as inorganic solids having specific gravity about 2.65 and organic solids having specific gravity 1.04. The particles having specific gravity more than 1.20 readily settle down at the bottom of the tank due to the force of gravity. This phenomenon of settlement is known as *hydraulic subsidence*. Every particle has its own hydraulic subsidence value. But the lighter the particle cannot settle down due to the force of gravity. Such particles are converted to settleable size by the application of some coagulant in water.

The process of settlement depends upon,

- a) Velocity of flow
- b) Size and shape of particles
- c) Viscosity of water.

Purpose of Sedimentation

- a) To reduce the heavy sediment load before the water enters the coagulation tank.
- b) To make the process of coagulation more easy.
- c) To reduce the amount of coagulant.
- d) To reduce the cost of coagulation process.
- e) To make the working of other treatment processes more efficient.

Coagulation and Flocculation

Coagulation

- Floc Formation: When coagulants are mixed with water thoroughly, a thick gelatinous precipitate is formed which is known as *floc*. This floc has got the property of attracting the suspended impurities in the water and settle down towards the bottom of tank.
- Electric charge : it is observed that the ions of floc possess positive electric charge and the colloidal particles possess negative electric charge so, the floc attracts the colloidal particles while it travels towards the bottom of tank.

Flocculation

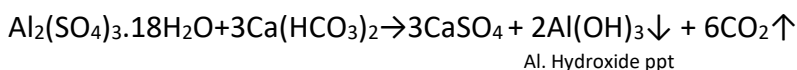
The phenomenon of formation of floc is termed as flocculation. The efficiency of flocculation is depend on the following factor.

- Dosing of coagulant.
- Mixing.
- pH-value

Chemicals used for Coagulation

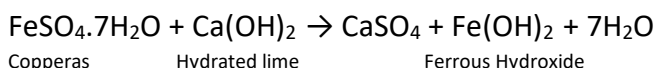
1. Use of Alum as coagulant ($\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$)

The alum when added to raw water, reacts with the bicarbonate alkalinities, which are generally present in raw supplies, so as to form a gelatinous precipitate (floc) of aluminium hydroxide. Floc attracts other fine particles and suspended matter (colloids), and thus grown in size, and finally settle down.

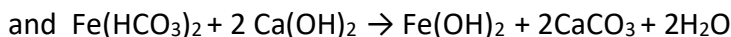
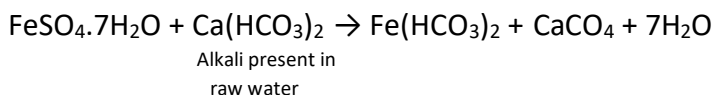


2. Use of Copperas as Coagulant. ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)

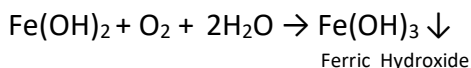
Copperas is generally added to raw water in conjunction with lime. Lime may be added either to copperas or vice versa. When lime added first, the following reaction takes place:



When copperas added first, the following reaction takes place :

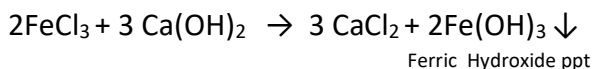
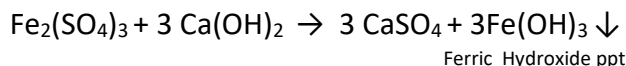


ferrous hydroxide further gets oxidized, formation of ferric hydroxide as given below,



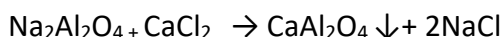
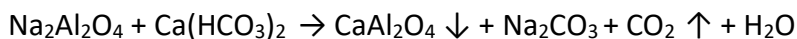
3. Use of Chlorinated Copperas as Coagulant.

When chlorine is added to a solution of copperas (i.e., ferrous sulphate), the two react chemically, so as to form ferric sulphate and ferric chloride. Chemical equation as follows:



4. Use of Sodium Aluminate as a Coagulant. ($\text{Na}_2\text{Al}_2\text{O}_4$)

This chemical when dissolved and mixed with water, reacts with the salts of calcium and magnesium present in raw water, resulting in the formation of precipitates of calcium or magnesium aluminate.



FILTRATION OF WATER

Theory of Filtration

The process of filtration consist in allowing the water to pass through the bed of sand. Thus, the colloidal impurities and some bacteria are removed. The theory of filtration is explain by the following phenomenon.

- a) Mechanical straining.
- b) Sedimentation
- c) Biological action
- d) Electrolytic action.

Classification of Filter.

- a) **Gravity filters** In these type of filters, the water percolates under the force of gravity. Again these filters may be of two type
 - i. **Slow Sand Filter.**
 - ii. **Rapid sand filter.**
- b) **Pressure filter** In this filter, the water is allowed to pass under a pressure grater than atmospheric pressure through a closed cylinder. Here the force of gravity has no function.

Slow Sand Filter

The theory of slow sand filter is based on the principle that if water is allowed to percolate slowly through the filtering media, then the biological, chemical and physical characteristic of water are improved considerably. And it permit sufficient time for those improvements. That's why the water is allowed to enter the filter bed slowly by suitable inlet arrangement. As the filtration takes much time it is not suitable for large scale, it is suitable for drinking water only for small towns.

Construction Feature

- a. **Enclosure Tank:-** It is a rectangular water tight tank constructed in brick masonry. The depth of the tank varies from 2-3.5m. The surface area depends on the volume of water to be filtered. Generally the surface area varies from 100m^2 - 2000m^2 .
- b. **Underdrainage system :-** It consist of a central drain and lateral drains. The central drain is a larger diameter pipe which extend from inlet side to outlet side. The lateral drain consist of perforated pips of smaller diameter which are connected to the central drain from both side.
- c. **Base Materials:-** The clean gravels of different sizes are used as base materials. It is placed over the under drainage system in four layer. Each layer is of 15 cm thickness. The bottom layer is made of bigger size gravels i.e. 40-65 mm. two intermediate layer are provided. In the first layer, the gravel size is 20-40 mm and in second layer, the gravel size is 6-20 mm. the top layer is made of smaller size gravel 3-6 mm.
- d. **Filter media of sand:-** The fine sand of effective size of 0.20-0.35 mm and uniformity co efficient 2-2.75 is generally used as the filtering media. The depth of the sand layer varies from 60-100 cm.
- e. **Appurtenance:-** The following appurtenance are provided:
 - i. A vertical air pipe is passed through the filter media for proper functioning of filter.
 - ii. A device for measuring loss of head.
 - iii. An adjustable telescopic tube to maintain constant discharge.

Rapid sand filter

It is observed that the rate of infiltration is more in coarse sand than that in fine sand. So, the theory of rapid sand filter is based on the principle of increasing the rate of filtration by providing coarse sand as filter media. The filtration head is also increased to increase the pressure head and the rate of filtration.

We have seen in slow sand filter that the rate of filtration is very low and it requires more space for installation. To overcome these difficulties, the fine sand in slow sand filter is replaced by coarse sand to achieve greater percolation and to reduce surface area.

Construction Feature

- a. **Enclosure tank :-** :- It consist of water tight tank constructed in brick masonry. The depth of the tank varies from 2-4 m. The surface area depends on the volume of water to be filtered. Generally the surface area varies from 30 - 60 m^2 .
- b. **Underdrainage system :-** It consist of a central drain and perforated lateral drains. The lateral drains are connected to the central drain from both sides and they are placed at a distance of 30 cm center to center.

- c. **Base materials:-** The clean gravels of different sizes are used as base materials. It is placed over the under drainage system in four layer. Each layer is of 15 cm thickness. The bottom layer is made of bigger size gravels i.e. 20-40 mm. two intermediate layer are provided. In the first layer, the gravel size is 12-20 mm and in second layer, the gravel size is 6-12 mm. The top layer is made of smaller size gravel 3-6 mm.
- d. **Filter media of sand :-** The coarse sand of effective size of 0.35-0.65 mm uniformity coefficient 1.20-1.80 is generally used as the filtering media. The depth of sand layer varies from 60-100 cm.
- e. **Appurtenances**
 - Air Compressor
 - Troughs
 - Rate control

DISINFECTION

NECESSITY OF DISINFACTION

The process of destroying harmful bacteria from water and making it safe for drinking is known as disinfection . The substance used for this purpose are known as disinfectants . The common disinfectants are lime, iodine and bromine , Ozone , potassium permanganate, silver chorine , etc, Chlorine is the most important disinfectant which has a wonderful power for killing bacteria in a short span of time with a minimum amount of expenditure .so, this chemical is used in most of the developing countries for the disinfection of water.

The necessity of disinfection may be described as follows:

- (a) After filtration, the water is found to carry some pathogenic bacteria which are responsible for water-borne diseases . Disinfection is necessary to destroy all such bacteria
- (b) In distribution system, her water may be contaminated by the leakage of pipe line or in some other way .Disinfection is system and up to the point of consumption.
- (c) Disinfection is necessary to protect the citizen from health hazard and to assure a healthy atmosphere to all.

METHODS OF DISINFECTION:

The following are the method of disinfection:

- I. **Disinfection by boiling:** When water is boiled to the boiling temp, the bacteria is completely removed. It should be boiled at least for 10-15 mins. It also removes some of the dissolved salts.

- II. **Disinfection by ultra-violet rays:** When mercury is enclosed in quartz bulbs and electric current passed through it, the UV rays are emitted, these rays are very powerful to kill all type of bacteria.
- III. **Disinfection by iodine and bromine:** It also kills bacteria. Dose of iodine or bromine should be 8-10 ppm. These chemicals are available in the form of small pallets. In this method, water is stored in suitable containers and required number of pallets are dropped in the containers and left for 5 mins.
- IV. **Disinfection by excess lime :** Normally, lime added to water to remove some of the dissolved salts. But when excess lime is added to water, it is found to act as disinfectant. The excess lime increase the pH value of water (i.e. increases alkalinity) which is detrimental to bacteria because the bacteria cannot resist the alkalinity of water. It is found that the pH value having 9-10 can remove bacteria to the extent of 99 percent. But, after treatment the residual lime should be removed by the method of recarbonation.
- V. **Disinfection by ozone:** In atmosphere, the molecule of oxygen contains two atoms (O_2). But it changes to three atoms when electric current under high voltage is passed through a stream of air in a chamber. This triatomic oxygen is known as ozone (O_3). The ozone easily breaks into oxygen (O_2) and nascent atom (O). this third atom is very powerful in killing bacteria. The dose of ozone varies from 2-5 ppm and contact period varies from 5-10 mins. If after treatment some residual ozone is present in water, it is automatically removed. Since ozone is unstable in nature, there is no possibility of any danger to the consumers.
It is a costly method and much care should be taken to avoid any accident due to electrical fault.
- VI. **Disinfection by potassium permanganate:** Potassium permanganate is a powerful oxidizing agent. It oxidizes the organic matters present in water and hence the bacteria are killed. But this is not suitable in large scale for public water supply schemes. This is mostly used for disinfecting the water of wells in village area, swimming pools, ponds, etc the dose of this chemical is about 2-3 ppm and the contact period is generally 2-3 hrs.
- VII. **Disinfection by silver:** Silver is found very effective in killing bacteria, silver foils are spread over the filter media and water is passed through it. The water absorbs some portion of silver which kills bacteria. The dose of silver varies from 0.5-1 ppm. As silver is costly, it is not suitable for public water supply schemes. It is suitable for domestic use only.
- VIII. **Disinfection by chlorine:** Chlorine has got the wonderful power of destroying bacteria. It is the best among all the other disinfectants used for the disinfection of water. It kills bacteria very fast and its effect lasts for such a long time that it even acts in exists in water, it does not cause any harm to consumers.

CHLORINATION

The process of disinfection of water by the application of chlorine is known as chlorination. The term chlorination indicates the destruction of bacteria from drinking waste water in order to protect the public health from water-borne.

APPLICATION OF CHLORINE

Chlorine may be applied in water in any one of the following form:

1. Chlorine
2. Chloramines
3. Bleaching powder
4. Chlorine dioxide

FORMS OF CHLORINATION

The following are the different forms of chlorination:

- 1) **Plain chlorination** : When the raw water is supplied to consumers by applying chlorine treatment only, it is termed as plain chlorination. In this case, the raw water is sufficiently clear and it does not require primary treatments like sedimentation, coagulation and filtration. Generally, in case of emergencies, for military camps, training camps, survey camps, etc. the process of plain chlorination is adopted.
- 2) **Pre- chlorination** : In rainy season, when the raw water is suspected to be highly contaminated then a dose of chlorine is added to raw water before it enters the sedimentation tank. This application of chlorine is termed as pre chlorination. That means, it is an advance dose of chlorine before the actual period of chlorination.
- 3) **Post- chlorination** : When chlorine is added to water after all the treatments are over and just before it enters the distribution system, it is termed as post chlorination. This dose of chlorine is added to water to control the contamination when it flows through the distribution system. Generally the dose of chlorine is about 1-2 ppm.
- 4) **Double chlorination** : When pre chlorination and post chlorination are both adopted in water supply scheme to safeguard the public health from epidemic, then it is termed as double chlorination. This system is urgently needed in case of flood and other natural calamity when the possibility of bacterial contamination are very high.
- 5) **Break Point chlorination** : It is a method of determining the chlorine demand of raw water.

There is no chlorine demand in pure water. If chlorine is added to such water, the chlorine will come as residual chlorine. When chlorine is added to contaminated water, it will kill the bacteria and oxidize the organic matters. In the beginning, the chlorine goes on killing

bacteria and then it starts accumulating up to a certain point . if the dose of chlorine goes on increasing, it will start emitting bad smell. But after sometime the bad smell suddenly disappears. This critical point is known as break point. If further chlorine is added it will come as residual chlorine. During the period of bad smell, the chlorine oxidizes the organic matters.

- 6) **Super-Chlorination:** The process of application of chlorine beyond the formation of beyond break point, is termed as super chlorination. This treatment is necessary when there is an epidemic due to some water-borne disease. But in this treatment much residual chlorine in water may appear which is unpleasant to consumers. To neutralize the residual chlorine to an allowable limit, dechlorination should be done.
- 7) **Dechlorination :** The process of removal of excess residual chlorine from water is known as dechlorination. The residual chlorine should not be removed completely, but some amount of chlorine about 0.25-0.5 ppm should remain in water so that it may not be polluted in the distribution system.

WATER SOFTENING

Water softening may be defined as the removal or reduction of hardness from water.

DEFINITION OF HARDNESS

The hardness of water is defined as the quality of water which is due to the presence of bicarbonate of calcium and magnesium; sulphates, chloride and nitrate of calcium and magnesium . such water is termed as hard water.

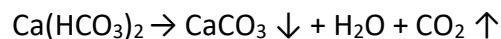
TYPES OF HARDNESS

Hardness may be two types:

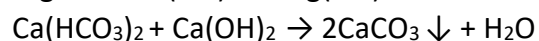
- (a) **Temporary hardness-** Temporary hardness is water hardness due to the presence of calcium and magnesium carbonates and bicarbonates, which can be precipitated by heating the water. It can be removed by processes such as boiling or lime softening, and then separation of water from the resulting precipitate.

Removal of temporary hardness by :

- **Boiling**

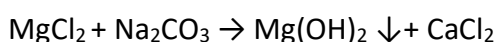
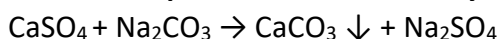


- **Adding lime**



(b) Permanent hardness- Permanent hardness in water is hardness due to the presence of the chlorides, nitrates and sulphates of calcium and magnesium, which will not be precipitated by boiling. The lime scale can build up on the inside of the pipe restricting the flow of water or causing a blockage.

Removal of permanent hardness by Lime Soda process



EFFECTS OF HARDNESS

The following are the effect of hardness:

- (a) It makes the food tasteless.
- (b) The vegetable, meat, etc. take much time to be boiled properly (i.e. soft)
- (c) It increase the fuel cost for cooking.
- (d) It consumes more soap and so it is uneconomical washing of clothes.
- (e) The working of dying system is highly affected as it causes the change of color shades in fabrics.
- (f) It reduce the life of fabrics.
- (g) It forms scale on boiler which reduce the life of steam engines.
- (h) It lead to corrosion and incrustation of pipes.

NECESSITY OF WATER SOFTENING

The softening of water is necessary for the following reasons:

- (a) To improve the taste of food.
- (b) To reduce the consumption of soap in washing of cloths.
- (c) To reduce the formation of scale in boilers.
- (d) To increase the life of fabrics.
- (e) To neutralize the effect on color in dying system.
- (f) To reduce the corrosive effect on pipes.